

Natural Computing

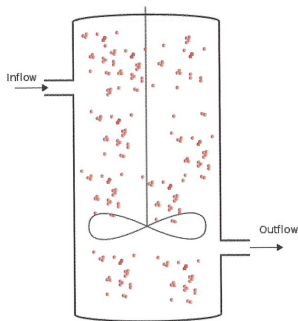


LUDWIG-
MAXIMILIANS-
UNIVERSITÄT
MÜNCHEN

LMU Munich
summer term 2025

where
is the

Thomas Gabor



Stirring Soup

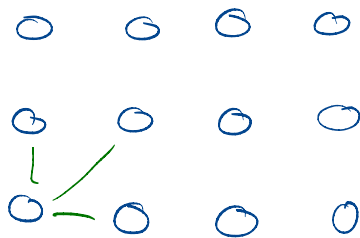


Figure 7.3 Well-stirred flow reactor with dilution — inflow and outflow of material.

well-stirred

ex:
reactor function A samples randomly
from reaction materials A

topological

ex:
reactor function A only chooses
reactions between neighbor particles

Stochasticity



No-Ops, Creation, and Destruction

$\emptyset \leftarrow$



$$A \rightarrow A + B$$

$$\emptyset \rightarrow C$$

$$A + B \rightarrow A$$

$$A \rightarrow \emptyset$$

example says:

$$X_{\epsilon} = \{\}$$

$$X_{\epsilon+1} = \{C\}$$

Example 4. A soup of functions writing into each other...?

Let $\mathcal{X} = \mathcal{X} \rightarrow \mathcal{X}$, i.e., the space of all functions that can take functions and return new functions.

Let $\mathcal{R} = \{x_1 + x_2 \rightarrow x_1 + x_3 \mid \underbrace{x_1, x_2, x_3 \in \mathcal{X}} \wedge \underbrace{x_3 = x_1(x_2)}\}$, i.e., when two functions x_1, x_2 meet, x_2 is overwritten with the result of applying x_1 to x_2 .

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

Let $X_0 \sim \wp^*(\mathcal{X})$.

Example: λ -calculus

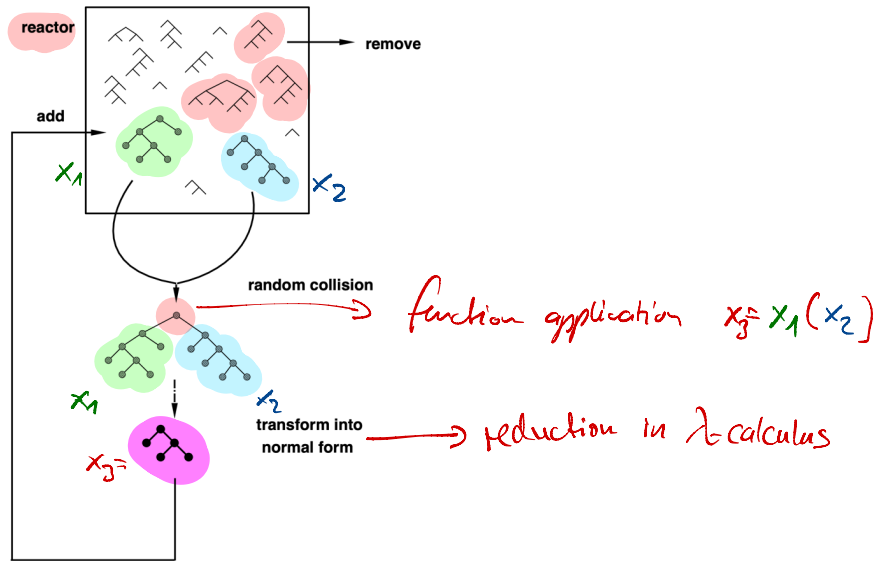


Figure 1: λ -calculus flow reactor. Two expressions A and B are chosen at random and a new object, $(A)B$, is constructed by “application” (see appendix A.2). Putting $(A)B$ into its normal form by β -reduction (see appendix A.2), effectively decides which object species (i.e., “stable molecular formula”) the new object is an instance of.

On λ -calculus...

Example:

$$(\lambda x. f \ x) \ y \quad \longrightarrow \quad f \ y$$

Example in Python:

$$(\text{lambda } x: \ f(x))(y) \quad \longrightarrow \quad f(y)$$

Example: λ -calculus

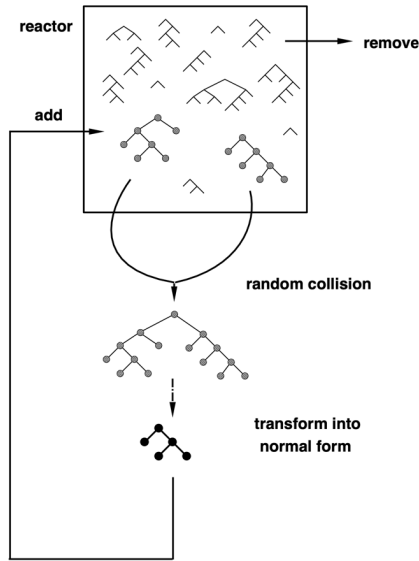


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Experiment Time??

Example 4. A soup of functions writing into each other...?

Let $\mathcal{X} = \underline{\mathcal{X}} \rightarrow \mathcal{X}$, i.e., the space of all functions that can take functions and return new functions.

Let $\mathcal{R} = \{\underline{x_1 + x_2} \mapsto \underline{x_1 + x_3} \mid x_1, x_2, x_3 \in \mathcal{X} \wedge x_3 = \underline{x_1}(\underline{x_2})\}$, i.e., when two functions x_1, x_2 meet, x_2 is overwritten with the result of applying x_1 to x_2 .

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

Let $\underline{X_0} \sim \wp^*(\mathcal{X})$.

$$\underline{x_1 = x_2} \mapsto \underline{(x \mapsto x)}$$

$$X_0 \subseteq \mathcal{X}$$

X_0

early

X_t

lates

X_t

very late

X_t

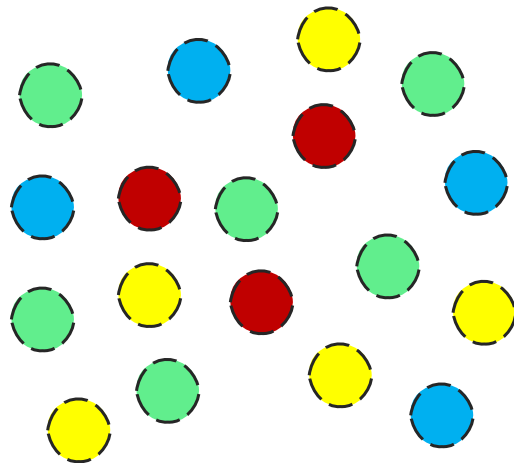
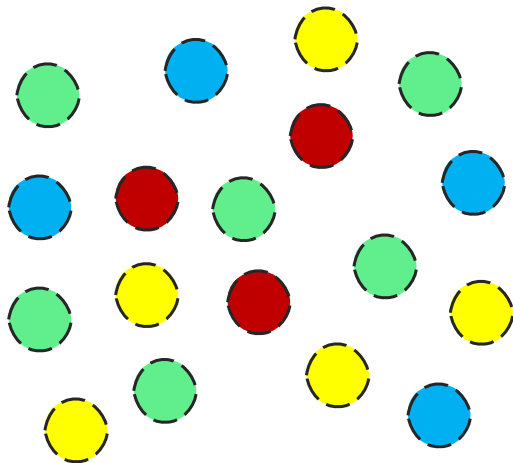
pretty chaotic behavior

still pretty chaotic, hardly predictable

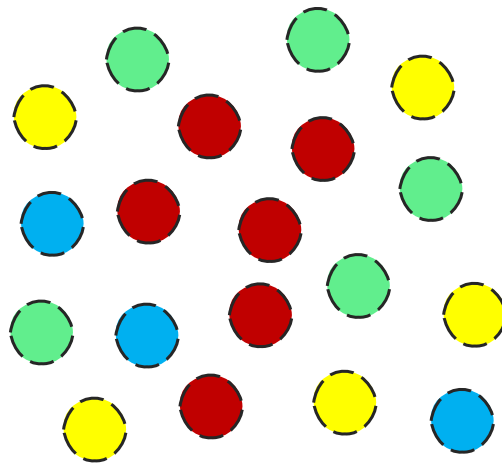
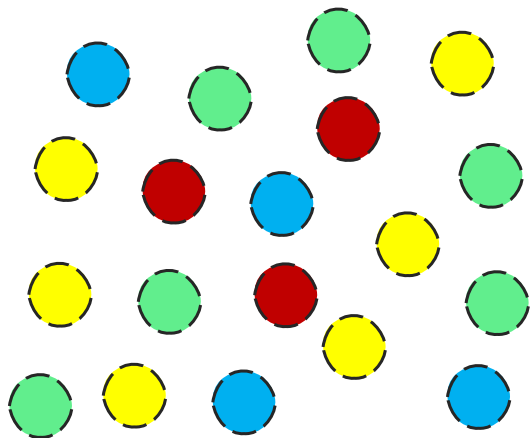
a little stability

replicative behavior $\rightarrow x_1 = x_2 \mapsto x_1$

Mere Stability vs. Self-Replication



Mere Stability vs. Self-Replication



Example: λ -calculus

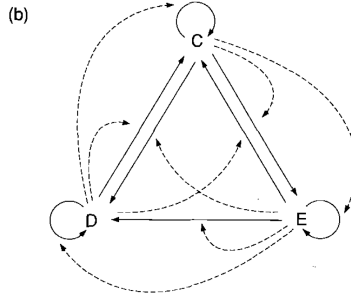
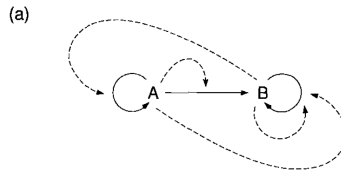


Figure 1. Level 0 ecology. A solid arrow denotes a transformation from argument object to value object. A dotted arrow connects the operator object with the transformation performed by it. Thus, $E \circ C = D$ is shown as a solid arrow from C to D pointed at by a dotted arrow originating in E .

$$\begin{aligned}
 A &\equiv \lambda x_1 . (x_1) \lambda x_2 . \lambda x_3 . (x_3) \lambda x_4 . \lambda x_5 . (x_5) x_4, & B &\equiv \lambda x_1 . (x_1) \lambda x_2 . \lambda x_3 . (x_3) x_2, \\
 C &\equiv \lambda x_1 . \lambda x_2 . (x_1) \lambda x_3 . x_1, & D &\equiv \lambda x_1 . \lambda x_2 . \lambda x_3 . (x_2) \lambda x_4 . x_2, \\
 E &\equiv \lambda x_1 . \lambda x_2 . \lambda x_3 . \lambda x_4 . (x_3) \lambda x_5 . x_3.
 \end{aligned}$$

Properties of Self-Replicators

longevity

fecundity

copy-fidelity

Levels of Organization

Complexity



Level 0: reproduction and ecology

- cycle of replicators (easily collapses to only one replicator)

Level 1: self-maintenance and organization

- emerges when mere copy actions are forbidden
- structure and interaction between particles can be described with specific rules
- part of the organization can generate all its elements anew

Level 2: organization of organizations

- create L1 organizations independently and then put them into the same soup
- organizations maintain themselves separately but interact via “glue” particles

Emergence of Implicit Target Functions

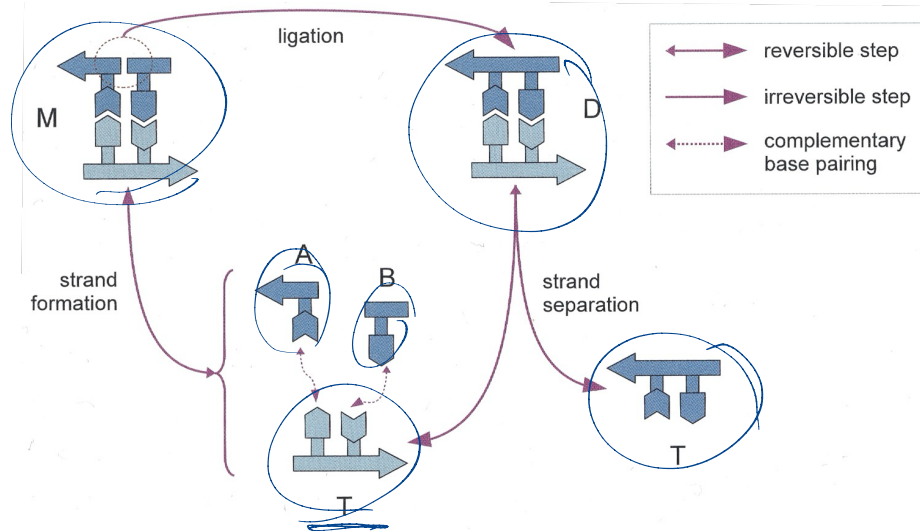


Figure 6.6 A model of the minimal self-replicator demonstrated by Von Kiedrowski [890]: The two *T* molecules are complementary templates (hexamers in the experiments). Each of them base-pairs with units *A* and *B* (trimers in the experiments), to form an intermediate molecule *M* which is almost a double strand: the still missing link between *A* and *B* is then catalyzed by *T* itself, forming the double strand *D*. *D* then splits into two single strands *T* and the cycle repeats. Adapted from [890].

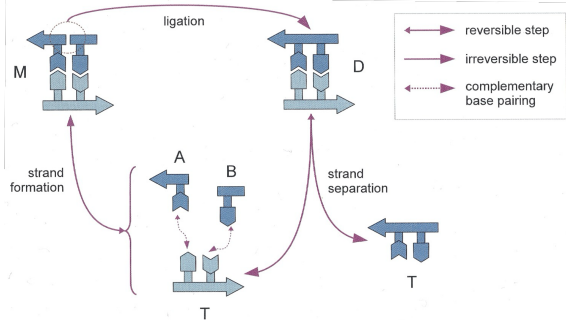


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Example 5. A DNA soup.

Let $\mathcal{X} = \{A, C, G, T\}^{2 \times 2} \cup \{A, C, G, T\}^2 \cup \{A, C, G, T\}^1$.

Let $\mathcal{R} = \{R_{1,-,-}, R_{2,-,-}, R_{3,-,-}, R_{4,-,-}\}$ with

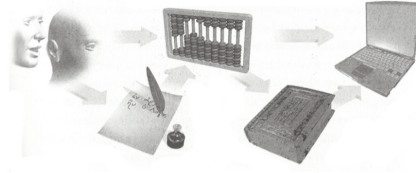
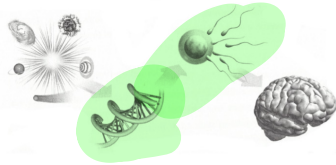
$$\begin{aligned}
 R_{1,X,Y} &= (X \ Y) + (-X) + (-Y) && \xrightarrow{0.5} \begin{pmatrix} X & Y \\ -X & -Y \end{pmatrix} \\
 R_{2,X,Y} &= \begin{pmatrix} X & Y \\ -X & -Y \end{pmatrix} && \xrightarrow{\quad} (X \ Y) + (-Y \ -X) \\
 R_{3,X,Y} &= (X) + (Y) && \xrightleftharpoons[0.01]{0.01} (X \ Y) \\
 R_{4,X} &= \emptyset && \xrightarrow{0.1} (X)
 \end{aligned}$$

for all $X, Y \in \{A, C, G, T\}$.

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

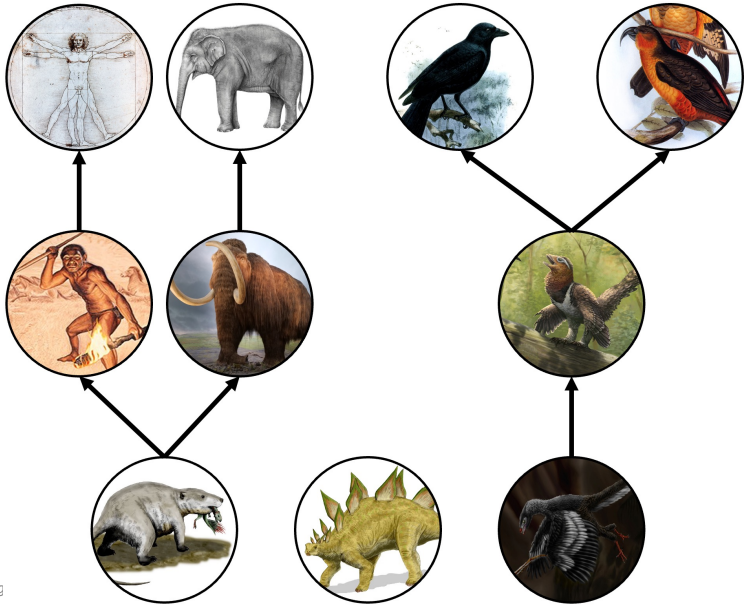
Let $X_0 = \emptyset$.

Natural Evolution



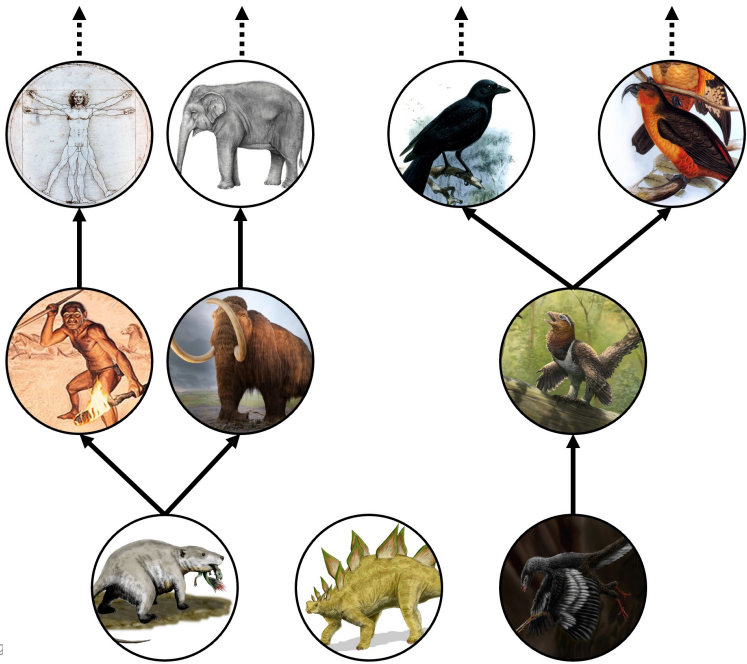
“The history of the universe can be thought of as a sequence of information processing revolutions, each of which builds on the technology of the previous ones.”

Seth Lloyd. Programming the Universe. Vintage Books, 2006.



de.wikipedia.org/wiki/Plesiosaurus#/media/Datei:Plesiosaurus_3DB.jpg
de.wikipedia.org/wiki/Stegosaurus#/media/Datei:Stegosaurus_BW.jpg
de.wikipedia.org/wiki/Evolution_der_Säugetiere#/media/Datei:Repenomamus_BW.jpg
de.wikipedia.org/wiki/Mammute#/media/Datei:Victoria_Mammut.P1033804.jpg
st.museum-digital.de/index.php?t=sammlung&gesusa=730
www.australiangeographic.com.au/topics/history-culture/2012/09/elephant-illustrations-the-art-of-science/
de.wikipedia.org/wiki/Vitruvianischer_Mensch#/media/Datei:Vitruvian_Man_by_Leonardo_da_Vinci.jpg
upload.wikimedia.org/wikipedia/commons/6/69/Balaur_bondoc_as_avialan.jpg
indianapublicmedia.org/amomentofscience/crow-thievery-2.php

Why does it keep going?



de.wikipedia.org/wiki/Plesiosaurus#/media/Datei:Plesiosaurus_3DB.jpg
de.wikipedia.org/wiki/Stegosaurus#/media/Datei:Stegosaurus_BW.jpg
de.wikipedia.org/wiki/Evolution_der_Säugetiere#/media/Datei:Repenomamus_BW.jpg
de.wikipedia.org/wiki/Mammute#/media/Datei:Victoria.Mammut.P1033804.jpg
st.museum-digital.de/index.php?t=sammlung&gesusa=730
www.australiangeographic.com.au/topics/history-culture/2012/09/elephant-illustrations-the-art-of-science/
de.wikipedia.org/wiki/Vitruvianischer_Mensch#/media/Datei:Vitruvian_Man_by_Leonardo_da_Vinci.jpg
upload.wikimedia.org/wikipedia/commons/6/69/Balaur_bondoc_as_avialan.jpg
indianapublicmedia.org/amomentofscience/crow-thievery-2.php

The Subject of Evolution

Individual
DNA

individual

group

Species



information in the genes

↳ information patterns in the gene pool

The Subject of Evolution

The Gene

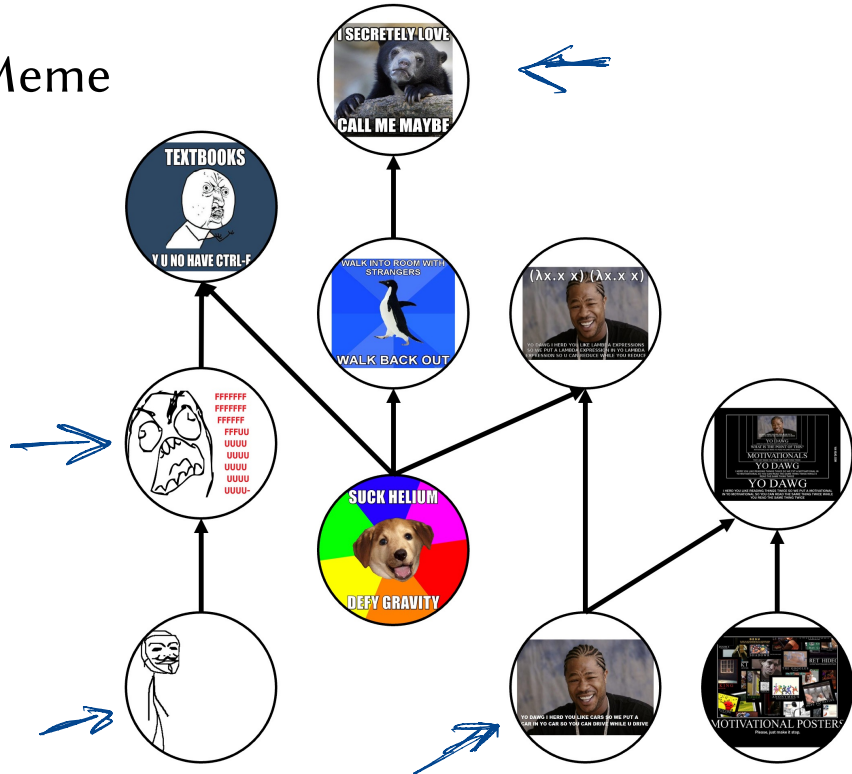
in the evolutionary biology sense

≈ information pattern

Other
(Non-Biological)
Instances of Evolution?

Detour: The Meme

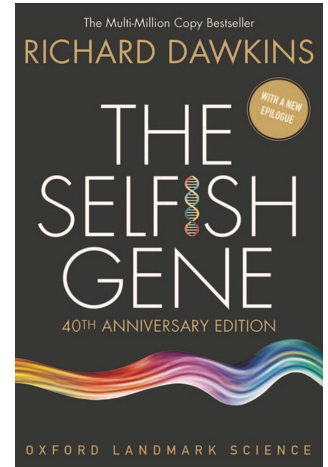
Detour: The (Internet) Meme



source: internet
also pt.slideshare.net/TjerkW/the-rise-of-functional-programming

Dawkins.
The Selfish Gene. (40th Anniversary Edition)
Oxford University Press, 2016.

ebookcentral.proquest.com/lib/ub-lmu/detail.action?docID=4545419



Evolutionary Computing

Definition 11 (population-based optimization). Let $\mathcal{X}$ be a state space. Let $\mathcal{T}$ be a target space with total order $\leq$. Let $\tau : \mathcal{X} \rightarrow \mathcal{T}$ be a target function. A tuple $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \tau, E, \langle X_t \rangle_{0 \leq t \leq g})$ is a population-based optimization process iff $X_t \in \wp^*(\mathcal{X})$ for all t and $E : \langle \wp^*(\mathcal{X}) \rangle \times (\mathcal{X} \rightarrow \mathcal{T}) \rightarrow \wp^*(\mathcal{X})$ is a possibly randomized, non-deterministic, or further parametrized function so that the population-based optimization run is produced by calling E repeatedly, i.e., $X_{t+1} = E(\langle X_u \rangle_{0 \leq u \leq t}, \tau)$ where X_0 is given externally or chosen randomly.

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Definition 12 (population-based optimization (alternate)). An optimization process $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \tau, E, \langle X_t \rangle_{0 \leq t \leq g})$ is called population-based iff $\mathcal{X}$ has the form $\mathcal{X} = \wp^*(\mathcal{Y})$ for some other state space $\mathcal{Y}$.

Algorithm 6 (basic evolutionary algorithm). Let $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \tau, E, \langle X_u \rangle_{0 \leq u \leq t})$ be a population-based optimization process. The process $\mathcal{E}$ continues via an evolutionary algorithm if E has the form

$$E(\langle X_u \rangle_{0 \leq u \leq t}, \tau) = X_{t+1} = \underbrace{\text{selection}(X_t \uplus \text{variation}(X_t))}$$

where *selection* and *variation* are possibly randomized or non-deterministic functions so that for any $X \in \wp^*(\mathcal{X})$ it holds that $|\text{selection}(X)| \leq |X|$ and $|\text{selection}(X \uplus \text{variation}(X))| = |X|$.